P25 Discriminator Capture Settings

Audacity recipe for raw C4FM discriminator audio from the BK M150, for GPS decoding (Tier 1 NMEA and peer-to-peer low-speed data). Version 1.0, 2026-11-07.

1. Windows input (once)

Setting	Value	Why
Input jack	**Line-in** (not mic)	Mic inputs add gain, bias and filtering
Sound settings > Input > device > Properties	Turn OFF all **enhancements**, AGC, noise suppression, echo cancel	These reshape the 4 levels
Advanced > Default format	**48000 Hz**, 16 or 24 bit	Matches the Audacity rate

2. Audacity settings

Setting	Value	Where
Host	**Windows WASAPI** (or MME if WASAPI won't show line-in)	Device toolbar
Recording device	The line-in the M150 discriminator feeds	Device toolbar
Channels	**1 (Mono)**	Device toolbar
Project rate	**48000 Hz**	Bottom-left / Edit > Preferences > Quality
Default sample format	**32-bit float** (16-bit OK)	Preferences > Quality
Effects / Auto-gain	**None.** No normalize, noise reduction, EQ, compressor	Leave the recording untouched
Level	Peaks about **-6 dB**, never clipping (no red)	Set with the PC input slider; check the meter during a key-up
Export	File > Export > **Export as WAV**, WAV (Microsoft) 32-bit float or signed 16-bit PCM. **Never MP3/OGG.**	One file per capture

3. Each recording

• Start recording, **2 s of quiet**, then key up. Talk or hold for **3 s or more** (several superframes repeat the position), unkey, **2 s of quiet**, stop.
• Keep the receiver and transmitter apart enough not to overload (a few feet plus low power, or a distant radio).
• Wait for a **steady GPS lock** icon on the transmitting radio before each key-up.
• File name: **mode_spot_take.wav**, e.g. tier1_A_1.wav, lsd_B_2.wav.

4. Capture plan

#	Mode	Spot	Takes	Purpose
1	Tier 1 location (NMEA), clear channel	A	3-4	Prove the whole chain: readable $GP... text
2	Peer-to-peer / low-speed data, clear	A	3-4	Same position repeated: find the stable bits
3	Peer-to-peer / low-speed data, clear	B (a few hundred ft due N or E of A)	3-4	Changing bits reveal lat vs lon
4	Emergency button send (optional)	A	1-2	See which path it uses

5. Capture log (one line per file)

Read the position off the radio display exactly as shown, and note its format (DD.dddddd or DD MM.mmm).

File name	Mode	Display lat	Display lon	Format	NAC	Unit ID	Dest ID	Notes

Send the WAV files plus this log (a photo of it is fine). The decoder checks every decoded field against these known values.